Chapter 2

Biological Classification

Since the dawn of civilisation there have been many attempts to classify living organisms, made instinctively rather than on scientific criteria, out of a need to use organisms for our own food, shelter and clothing.

Aristotle, the earliest scientific attempt

- used simple morphological characters
- classified plants into trees, shrubs and herbs
- divided animals into those which had red blood and those that did not

The Two Kingdom system

- developed in Linnaeus' time, with Plantae and Animalia taking in all plants and all animals
- it did not distinguish eukaryotes from prokaryotes, unicellular from multicellular, or photosynthetic green algae from non-photosynthetic fungi
- sorting organisms into plants and animals was easy to do and easy to understand, but a large number of organisms fell into neither category

Why the systems kept changing

- used for a long time, the two kingdom classification was found inadequate
- besides gross morphology, a need was felt to include cell structure, nature of wall, mode of nutrition, habitat, methods of reproduction and evolutionary relationships
- plant and animal kingdoms have been a constant under every system, but which organisms belong in them has kept changing, and the number and nature of the other kingdoms has been understood differently by different scientists over time

In this chapter

2.2 Kingdom Monera

2.4 Kingdom Protista

2.6 Kingdom Fungi

2.9 Kingdom Plantae

2.10 Kingdom Animalia

2.11 Viruses, viroids, prions and lichens

2.1 The five kingdom system

Whittaker's five kingdoms

- R.H. Whittaker (1969) proposed a Five Kingdom Classification
- the kingdoms he defined were Monera, Protista, Fungi, Plantae and Animalia
- his main criteria were cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships

A three domain system as well

- a three-domain system has also been proposed
- it divides Kingdom Monera into two domains and leaves the remaining eukaryotic kingdoms in the third, giving a six kingdom classification
- that system is studied in detail in higher classes

What the earlier systems lumped together

- bacteria, blue green algae, fungi, mosses, ferns, gymnosperms and angiosperms all sat under Plants, unified only by every organism in the group having a cell wall
- that placed prokaryotic bacteria and blue green algae (cyanobacteria) beside eukaryotes, and unicellular organisms beside multicellular ones, with Chlamydomonas and Spirogyra together under algae
- it did not separate the heterotrophic fungi from the autotrophic green plants, though their wall composition differed too, the fungi having chitin

Table 2.A Characteristics of the Five Kingdoms

Characters	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present with chitin	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular / loose tissue	Tissue / organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic) and Heterotrophic	Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic)	Heterotrophic (holozoic / saprophytic etc.)

Cocci, spherical

Bacilli, rod-shaped

Vibrio, comma-shaped

Spirilla, spiral

Flagellum, the motile thread some forms carry

Spore, the resting body formed in unfavourable conditions

Figure 2.A Bacteria of different shapes

Why fungi and protists were split off

- the fungi had chitin in their walls while the green plants had a cellulosic cell wall
- once such characters were weighed, fungi were placed in a separate Kingdom Fungi
- all prokaryotic organisms were grouped under Kingdom Monera, and the unicellular eukaryotes under Kingdom Protista

What Protista brought together

- Chlamydomonas and Chlorella, earlier placed in Algae within Plants, both having cell walls
- Paramoecium and Amoeba, earlier in the animal kingdom, which lack a cell wall
- organisms from different kingdoms came together because the criteria for classification changed, and such changes will happen again as our understanding of characters and evolutionary relationships improves

The aim has been a system that reflects morphological, physiological and reproductive similarities and is also phylogenetic, that is, based on evolutionary relationships. Monera, Protista and Fungi are studied here; Plantae and Animalia are dealt with separately in chapters 3 and 4.

2.2 Kingdom Monera

Bacteria, the sole members

- bacteria are the sole members of Kingdom Monera and the most abundant micro-organisms
- hundreds are present in a handful of soil and they occur almost everywhere
- they live in extreme habitats such as hot springs, deserts, snow and deep oceans, where very few other life forms survive
- many live in or on other organisms as parasites
- they are grouped in four categories by shape, shown in Figure 2.A

Filament, cells joined end to end in a chain

Heterocyst, the specialised cell in which atmospheric nitrogen is fixed

Gelatinous sheath, surrounding the colony

Figure 2.B A filamentous blue-green algae – Nostoc

Simple in structure, complex in behaviour

- as a group bacteria show the most extensive metabolic diversity of any organisms
- some are autotrophic and synthesise their own food from inorganic substrates, either photosynthetically or chemosynthetically
- the vast majority are heterotrophs, depending on other organisms or on dead organic matter

2.3 Archaebacteria and eubacteria

Archaebacteria

- live in some of the harshest habitats: extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens)
- a different cell wall structure is what lets them survive extreme conditions
- methanogens live in the gut of ruminants such as cows and buffaloes and produce methane, or biogas, from their dung

Eubacteria, the true bacteria

- thousands of them, marked by a rigid cell wall and, if motile, a flagellum
- cyanobacteria, also called blue-green algae, have chlorophyll a like green plants and are photosynthetic autotrophs
- they are unicellular, colonial or filamentous, and freshwater, marine or terrestrial
- colonies are generally surrounded by a gelatinous sheath, and they often form blooms in polluted water bodies

Fixing nitrogen, and chemosynthesis

- some fix atmospheric nitrogen in specialised cells called heterocysts, for example Nostoc and Anabaena
- chemosynthetic autotrophic bacteria oxidise inorganic substances such as nitrates, nitrites and ammonia, using the released energy for ATP production
- they play a great role in recycling nutrients such as nitrogen, phosphorous, iron and sulphur

Heterotrophic bacteria

- the most abundant in nature, and the majority are important decomposers
- many matter to human affairs: making curd from milk, producing antibiotics, and fixing nitrogen in legume roots

Bacterial cell, constricted across the middle

Fission, the single cell separating into two daughter cells

This is the main means by which bacteria reproduce

Figure 2.C A dividing bacterium

Bacteria that cause disease, and how they reproduce

- some are pathogens damaging human beings, crops, farm animals and pets
- cholera, typhoid, tetanus and citrus canker are well known bacterial diseases
- they reproduce mainly by fission, and under unfavourable conditions produce spores
- they also carry out a sort of sexual reproduction, a primitive transfer of DNA from one bacterium to another

Mycoplasma

- completely lack a cell wall
- the smallest living cells known
- can survive without oxygen, and many are pathogenic in animals and plants

2.4 Kingdom Protista

All single-celled eukaryotes

- the boundaries of this kingdom are not well defined; what is a photosynthetic protistan to one biologist may be a plant to another
- chrysophytes, dinoflagellates, euglenoids, slime moulds and protozoans are included here
- members are primarily aquatic, and the kingdom forms a link with plants, animals and fungi
- being eukaryotes, the cell body has a well defined nucleus and other membrane-bound organelles, and some have flagella or cilia
- they reproduce asexually and sexually, by a process involving cell fusion and zygote formation

Dinoflagellates, stiff cellulose plates on the wall and two flagella, one longitudinal and one transverse

Euglena, a pellicle in place of a cell wall, with one short and one long flagellum

Slime mould, the plasmodium that creeps over decaying twigs and leaves

Paramoecium, a protozoan of the ciliated group

Figure 2.D (a) Dinoflagellates (b) Euglena (c) Slime mould (d) Paramoecium

2.5 The protistan groups

Chrysophytes

- diatoms and golden algae (desmids), of fresh water and marine environments; microscopic, floating passively as plankton, mostly photosynthetic
- in diatoms the walls form two thin overlapping shells fitting as in a soap box, embedded with silica and so indestructible
- their deposits accumulated over billions of years as diatomaceous earth, gritty soil used in polishing and in filtering oils and syrups
- diatoms are the chief producers in the oceans

Dinoflagellates

- mostly marine and photosynthetic, appearing yellow, green, brown, blue or red by the main pigments in their cells
- the cell wall carries stiff cellulose plates on the outer surface
- most have two flagella, one longitudinal, the other transverse in a furrow between the plates
- red forms such as Gonyaulax multiply so fast the sea appears red, and their toxins may kill marine animals such as fishes

Euglenoids

- mostly fresh water organisms of stagnant water, with two flagella, one short and one long
- instead of a cell wall a protein rich layer called pellicle, which makes the body flexible
- photosynthetic in sunlight, but deprived of it they behave as heterotrophs, predating on smaller organisms
- their pigments are identical to those of higher plants

Slime moulds

- saprophytic protists whose body moves along decaying twigs and leaves, engulfing organic material
- in suitable conditions they form an aggregation called plasmodium, which may spread over several feet
- in unfavourable conditions it differentiates into fruiting bodies bearing spores at their tips; the spores have true walls, resist extremes, survive years and travel on air currents

Protozoans

- all are heterotrophs living as predators or parasites
- believed to be primitive relatives of animals
- there are four major groups
- amoeboid protozoans live in fresh water, sea water or moist soil

Amoeboid and flagellated protozoans

- amoeboid forms move and capture prey by putting out pseudopodia, or false feet, as in Amoeba
- marine amoeboid forms have silica shells on their surface, and some, such as Entamoeba, are parasites
- flagellated protozoans are either free-living or parasitic and carry flagella
- the parasitic flagellated forms cause diseases such as sleeping sickness, for example Trypanosoma

Ciliated and sporozoan protozoans

- ciliated forms are aquatic and actively moving, because of thousands of cilia
- they have a cavity, the gullet, opening to the outside of the cell surface, and coordinated movement of rows of cilia steers water laden with food into it, as in Paramoecium (Figure 2.D)
- sporozoans are diverse organisms with an infectious spore-like stage in the life cycle
- the most notorious is Plasmodium, the malarial parasite, which causes malaria, a disease with a staggering effect on human population

2.6 Kingdom Fungi

A unique kingdom of heterotrophs

- great diversity in morphology and habitat: on moist bread and rotten fruits, the common mushroom and toadstools, white spots on mustard leaves from a parasitic fungus
- some unicellular fungi such as yeast are used to make bread and beer
- others cause disease in plants and animals, the wheat rust-causing Puccinia being an important example, and some are a source of antibiotics, such as Penicillium
- they are cosmopolitan, occurring in air, water, soil and on animals and plants, preferring warm and humid places, which is why food is kept in the refrigerator

The fungal body

- with the exception of the unicellular yeasts, fungi are filamentous
- their bodies consist of long, slender thread-like structures called hyphae, and the network of hyphae is the mycelium
- some hyphae are continuous tubes filled with multinucleated cytoplasm, called coenocytic; others have septae or cross walls
- the cell walls are composed of chitin and polysaccharides

How fungi feed and reproduce

- most are heterotrophic and absorb soluble organic matter from dead substrates, and so are saprophytes
- those depending on living plants and animals are parasites
- they can also live as symbionts, with algae as lichens and with the roots of higher plants as mycorrhiza
- vegetative reproduction is by fragmentation, fission and budding, and asexual reproduction is by spores

Mucor, a phycomycete, with aseptate coenocytic mycelium

Aspergillus, an ascomycete, bearing conidia on conidiophores

Agaricus, a basidiomycete, the mushroom, with its basidiocarp

Figure 2.E Fungi: (a) Mucor (b) Aspergillus (c) Agaricus

Spores and the sexual cycle

- asexual spores are called conidia, sporangiospores or zoospores; sexual spores are oospores, ascospores and basidiospores
- the various spores are produced in distinct structures called fruiting bodies
- the sexual cycle runs in three steps: plasmogamy, the fusion of protoplasms between two motile or non-motile gametes; karyogamy, the fusion of two nuclei; and meiosis in the zygote, giving haploid spores

The dikaryophase

- when a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse
- in some fungi the fusion of two haploid cells at once gives diploid cells (2n)
- in ascomycetes and basidiomycetes an intervening dikaryotic stage occurs, with two nuclei per cell (n + n); this condition is a dikaryon and the phase the dikaryophase
- later the parental nuclei fuse, the cells become diploid, and reduction division in the fruiting bodies forms haploid spores

2.7 Phycomycetes and ascomycetes

Phycomycetes

- found in aquatic habitats, on decaying wood in moist damp places, or as obligate parasites on plants
- the mycelium is aseptate and coenocytic
- asexual reproduction is by motile zoospores or non-motile aplanospores, produced endogenously in a sporangium
- a zygospore forms by fusion of two gametes, similar in morphology (isogamous) or dissimilar (anisogamous or oogamous)
- examples are Mucor, Rhizopus (the bread mould) and Albugo (the parasitic fungi on mustard)

The morphology of the mycelium, the mode of spore formation and the fruiting bodies form the basis for dividing the kingdom into classes. Ascomycetes, the sac-fungi, are mostly multicellular, such as Penicillium, or rarely unicellular, such as yeast (Saccharomyces); they are saprophytic, decomposers, parasitic or coprophilous.

Ascomycetes, concluded

- the mycelium is branched and septate
- asexual spores are conidia, produced exogenously on special mycelium called conidiophores, and conidia on germination produce mycelium
- sexual spores are ascospores, produced endogenously in sac like asci (singular ascus), and the asci are arranged in fruiting bodies called ascocarps
- examples are Aspergillus (Figure 2.E), Claviceps and Neurospora, the last used extensively in biochemical and genetic work; morels and truffles are edible and considered delicacies

2.8 Basidiomycetes and deuteromycetes

Basidiomycetes

- commonly known as mushrooms, bracket fungi or puffballs
- they grow in soil, on logs and tree stumps, and in living plant bodies as parasites, such as rusts and smuts; the mycelium is branched and septate
- asexual spores are generally not found, but vegetative reproduction by fragmentation is common
- sex organs are absent, and plasmogamy comes about by fusion of two vegetative or somatic cells of different strains or genotypes
- the resultant dikaryotic structure gives rise to a basidium, where karyogamy and meiosis produce four basidiospores exogenously; basidia sit in fruiting bodies called basidiocarps, and common members are Agaricus the mushroom (Figure 2.E), Ustilago the smut and Puccinia the rust fungus

Deuteromycetes

- called imperfect fungi, because only the asexual or vegetative phases are known
- when the sexual forms were discovered these fungi were moved into the classes they rightly belong to, often ascomycetes and basidiomycetes
- an asexual stage and a sexual stage of the same fungus could be given separate names and placed in separate classes, until the linkage was established
- they reproduce only by asexual spores known as conidia, and the mycelium is septate and branched
- some are saprophytes or parasites, while a large number are decomposers of litter that help in mineral cycling; examples are Alternaria, Colletotrichum and Trichoderma

2.9 Kingdom Plantae

What Plantae takes in

- all eukaryotic chlorophyll-containing organisms, commonly called plants
- a few members are partially heterotrophic, such as the insectivorous bladderwort and Venus fly trap, and the parasite Cuscuta
- plant cells have a eukaryotic structure with prominent chloroplasts and a cell wall mainly of cellulose
- algae, bryophytes, pteridophytes, gymnosperms and angiosperms are included

Alternation of generations

- the life cycle has two distinct phases, the diploid sporophytic and the haploid gametophytic, which alternate with each other
- the lengths of the haploid and diploid phases, and whether they are free-living or dependent on others, vary among groups of plants
- this phenomenon is called alternation of generations

2.10 Kingdom Animalia

Heterotrophic eukaryotes without cell walls

- multicellular, and their cells lack cell walls
- they depend directly or indirectly on plants for food, digest it in an internal cavity, and store reserves as glycogen or fat
- their mode of nutrition is holozoic, by ingestion of food
- they follow a definite growth pattern and grow into adults of definite shape and size; higher forms show elaborate sensory and neuromotor mechanism and most can locomote
- sexual reproduction is by copulation of male and female, followed by embryological development

2.11 Viruses, viroids, prions and lichens

Why viruses fall outside the five kingdoms

- Whittaker's classification makes no mention of lichens or of acellular organisms such as viruses, viroids and prions
- viruses found no place because they are not considered truly living, if living means having a cell structure
- they are non-cellular organisms marked by an inert crystalline structure outside the living cell

Tobacco Mosaic Virus, a rod of capsid protein around the nucleic acid

Bacteriophage, head and collar above, sheath and tail fibres below

Capsid, the protein coat of capsomeres that protects the nucleic acid

RNA, the genetic material carried inside

Figure 2.F (a) Tobacco Mosaic Virus (TMV) (b) Bacteriophage

How viruses were found

- once they infect a cell they take over the host machinery to replicate themselves, killing the host
- virus means venom or poisonous fluid
- Dmitri Ivanowsky (1892) recognised certain microbes as the causal organism of tobacco mosaic disease, smaller than bacteria because they passed through bacteria-proof filters
- M.W. Beijerinck (1898) showed extract of infected tobacco could infect healthy plants, named the pathogen virus and called the fluid Contagium vivum fluidum, infectious living fluid
- W.M. Stanley (1935) showed viruses could be crystallised and that the crystals consist largely of proteins

Genetic material

- besides proteins, viruses contain genetic material, either RNA or DNA, never both
- a virus is a nucleoprotein and its genetic material is infectious
- viruses infecting plants generally have single stranded RNA; those infecting animals have single or double stranded RNA or double stranded DNA
- bacteriophages, the viruses that infect bacteria, are usually double stranded DNA viruses

Structure and disease

- the protein coat, the capsid, is made of small subunits called capsomeres, which protect the nucleic acid
- capsomeres are arranged in helical or polyhedral geometric forms
- viruses are inert outside their specific host cell and are obligate parasites
- they cause mumps, small pox, herpes and influenza, and AIDS in humans is also caused by a virus

Symptoms in plants

- mosaic formation, leaf rolling and curling
- yellowing and vein clearing
- dwarfing and stunted growth

Viroids

- in 1971 T.O. Diener discovered a new infectious agent smaller than viruses, causing potato spindle tuber disease
- it was found to be a free RNA, lacking the protein coat found in viruses, hence the name viroid
- the RNA of the viroid was of low molecular weight

Prions

- certain infectious neurological diseases were found to be transmitted by an agent consisting of abnormally folded protein, similar in size to viruses
- these agents were called prions
- the most notable prion diseases are bovine spongiform encephalopathy (BSE), called mad cow disease in cattle, and its analogous variant Creutzfeldt–Jacob disease (CJD) in humans

Lichens

- symbiotic, that is mutually useful, associations between algae and fungi
- the algal component is the phycobiont and the fungal component the mycobiont, autotrophic and heterotrophic respectively
- algae prepare food for the fungi, and fungi provide shelter and absorb mineral nutrients and water for their partner
- so close is the association that one would never imagine two different organisms within a lichen; they are very good pollution indicators and do not grow in polluted areas

2.12 Summary

The systems, and the five kingdoms

- biological classification of plants and animals was first proposed by Aristotle on simple morphological characters, and Linnaeus later placed all living organisms in Plantae and Animalia
- Whittaker proposed the elaborate five kingdom classification of Monera, Protista, Fungi, Plantae and Animalia
- its main criteria were cell structure, body organisation, mode of nutrition and reproduction, and phylogenetic relationships

Kingdom by kingdom

- bacteria fall in Monera, are cosmopolitan, show the most extensive metabolic diversity and may be autotrophic or heterotrophic
- Protista holds all single-celled eukaryotes, with a defined nucleus and membrane bound organelles, reproducing both asexually and sexually
- Fungi show great diversity in structure and habitat, are mostly saprophytic, and fall into phycomycetes, ascomycetes, basidiomycetes and deuteromycetes
- Plantae holds the eukaryotic chlorophyll-containing organisms and shows alternation of generations; Animalia holds the heterotrophic, multicellular eukaryotes lacking a cell wall

2.13 Exercises

From this chapter

1. Discuss how classification systems have undergone several changes over a period of time.
2. State two economically important uses of: (a) heterotrophic bacteria (b) archaebacteria
3. What is the nature of cell walls in diatoms?
4. Find out what the terms algal bloom and red tides signify.
5. How are viroids different from viruses?
6. Describe briefly the four major groups of Protozoa.
7. Plants are autotrophic. Can you think of some plants that are partially heterotrophic?
8. What do the terms phycobiont and mycobiont signify?
9. Give a comparative account of the classes of Kingdom Fungi under the following: (i) mode of nutrition (ii) mode of reproduction
10. What are the characteristic features of Euglenoids?
11. Give a brief account of viruses with respect to their structure and nature of genetic material. Also name four common viral diseases.
12. Organise a discussion in your class on the topic: are viruses living or non-living?